

VetMyPet - AI Animal Health & Diagnostic Platform

An End-to-End Multimodal Video, Vision & Clinical Decision Support System

Platform Version v2.5.0 (Production Core)

Core Backend Go 1.22+ / SQLite WAL

Clinical AI Model viggovet/viggoVet-Reasoning-20B

Generated Date August 2026

1. Executive Summary & System Overview

The **VetMyPet - AI Animal Health & Diagnostic Platform** is a full-stack, multimodal clinical decision support system engineered for veterinary practitioners, veterinary technicians, agricultural livestock managers, and companion animal owners. The system ingests high-resolution still photographs and temporal live video recordings (e.g. motion, involuntary spasms, chorea, gait abnormalities, respiratory distress), extracts sequential keyframes, conducts objective computer vision inspection, and synthesizes visual evidence with owner-reported clinical history using specialized Large Multimodal Models (LMMs).

Unlike conventional veterinary AI classifiers that rigidly classify dermatology lesions or force single

disease predictions, this platform operates on an **unbiased, broad-spectrum differential diagnosis framework**. It evaluates infectious (viral, bacterial, fungal, parasitic), non-infectious (mechanical obstructions, toxicological, inflammatory, metabolic, traumatic), and behavioral presentations across canine, feline, bovine, equine, caprine, ovine, and avian species using `viggovet/viggoVet-Reasoning-20B` for veterinary clinical reasoning.

2. Problem Statement & Key Industry Challenges

Veterinary triage and diagnostics face significant operational and technical hurdles:

- **The Static Image Diagnostic Blindspot:** Many life-threatening veterinary conditions cannot be diagnosed from a single static photograph. For example, *Canine Distemper Virus (CDV)* presents with rhythmic chorea/myoclonus (jaw and leg twitching), *Equine Colic* presents with dynamic visceral pain (ground pawing, flank watching, rolling), and *Rabies* presents with behavioral agitation or ataxia. These require motion analysis across temporal video frames.
- **Classifier Bias & False Positives:** Traditional image classification models are trained on narrow dermatological datasets, frequently misclassifying critical systemic emergency conditions (e.g. Parvovirus, Distemper, AHDS) as superficial skin pyoderma or forcing predetermined diseases.
- **Data Integrity & Concurrency Lockouts:** In embedded database deployments (e.g. SQLite in veterinary clinics and field laptops), continuous automated snapshot backups frequently trigger `database is locked` errors during write-heavy clinical screening workflows.
- **Auth Header & API Token Collisions:** In multi-tier architectures, forwarding user login bearer tokens into third-party AI provider headers triggers HTTP 401/403 authorization rejections.

3. Our Solution & Core Innovations



Multimodal Video & Keyframe Pipeline

In-browser camera recording (MediaRecorder) and file upload (.mp4, .webm, .mov) with automated ffmpeg

temporal sampling extracting 4 evenly-spaced keyframes across timeline adhering to AI router quotas.



Dual-Stage Multimodal AI Pipeline

Strict separation of concerns: Stage 1 (Qwen2.5-VL-72B-Instruct) performs purely objective visual/motion extraction; Stage 2 (viggovet/viggoVet-Reasoning-20B) executes specialized veterinary internal medicine reasoning.

Dual-Database & Point-in-Time Backups

Dedicated `user.db` (Bcrypt password hashing, token auth) and `animal_health.db` (records, media, screenings) with automatic `VACUUM INTO` point-in-time snapshot rotation upon every database mutation.

Lock-Free SQLite Concurrency

WAL mode with `_busy_timeout=10000`, `PRAGMA busy_timeout = 10000;;`, and serialized write pooling (`SetMaxOpenConns(1)`) completely eliminating lock contention during simultaneous auto-backups.

4. Complete Technology Stack & Specifications

Tier / Component	Technologies & Libraries	Functional Responsibility
Frontend Layer	React 18 Vite SPA Tailwind CSS Lucide Icons	Single-page web application featuring responsive dark/light interface, live camera video capture, drag-and-drop media upload, triage visualization, and patient records management.
Client APIs & Capture	MediaRecorder API Navigator.mediaDevices Fetch API	In-browser 5–12s gait/spasm video recording, live video preview playback, audio/video stream constraints, and custom header injection (<code>X-Auth-Token</code> , <code>X-User-ID</code>).
Backend Core	Go 1.22+ net/http golang.org/x/crypto/bcrypt	High-throughput REST API server, custom middleware pipeline (CORS, Token Auth, Request Logging), and

Tier / Component	Technologies & Libraries		Functional Responsibility
			asynchronous backup event dispatchers.
Media Engine	FFmpeg ffprobe	6.0+	Video stream validation, duration probing, temporal filter-graph scaling (<code>fps=1/2,scale=512:-1</code>), and 4-frame JPEG base64 extraction.
Databases & Storage	SQLite <code>mattn/go-sqlite3</code>	3 (WAL)	Two isolated databases: <code>user.db</code> (authentication, credentials, salt/hashes) and <code>animal_health.db</code> (patients, media metadata, clinical screenings, reminders).
AI Inference Layer	<code>Qwen2.5-VL-72B-Instruct</code> <code>viggovet/viggoVet-Reasoning-20B</code> Hugging Face Router API		Multimodal visual analysis on keyframe arrays, followed by structured veterinary clinical assessment, differential ranking, confirmatory test prescription, and triage urgency rating.

5. End-to-End System Architecture & Data Pipelines

+-----

-----+ | FRONTEND CLIENT (React 18 SPA) | | [Photo Capture / File Drag-Drop] [In-Browser Video

Recorder (MediaRecorder)] [Symptoms Form] | +-----

+-----+ | Multipart POST /api/health-check/upload

(Headers: X-Auth-Token, X-User-ID) v

+-----

-----+ | BACKEND INGESTION & STORAGE ENGINE | | 1. MIME & Size Validation

(.jpg, .png, .mp4, .webm, .mov) | | 2. Content-Addressable Storage Write -> ./uploads/{hash}.ext |

| 3. Register Media Record in animal_health.db (Type: 'video' | 'image') |

+-----

+-----+ | POST /api/health-check/analyze { media_id:

11, symptoms: [...] } v

+-----

-----+ | TEMPORAL KEYFRAME EXTRACTION (FFmpeg) | | If video -> ffmpeg -y -i input.webm -vf

"fps=1/2,scale=512:-1" -vframes 4 frame_%03d.jpg | | Collect 4 base64 data URLs:

["data:image/jpeg;base64,...", "data:image/jpeg;base64,..."] |

+-----

+-----+ | Payload: 4 Image Data URLs + Unbiased

Objective	Inspection	Prompt	v
+-----+ STAGE 1: MULTIMODAL COMPUTER VISION AI Model: Qwen/Qwen2.5-VL-72B-Instruct (Vision)			
Outputs JSON: species, posture, visible lesions, discharges, wounds, gait/chorea, behavior, severity			
+-----+ Payload: Species + Vision Findings +			
Clinical	Complaints	/	History
+-----+ STAGE 2: VETERINARY INTERNAL MEDICINE REASONING Model: viggovet/viggoVet-Reasoning-20B (Clinical Specialist) Outputs JSON: 4-5 ranked differentials, pathophysiological reasons, confirmatory tests, triage urgency			
+-----+ Persistence & Automated Backup Event Hook (OnChange)			
+-----+ DATABASE PERSISTENCE & ASYNC BACKUP ENGINE 1. Auto-resolve & persist Animal Record (linked to authenticated User) 2. Insert Health Screening Record in animal_health.db 3. Trigger OnChange() -> Asynchronous Snapshot Backup via VACUUM INTO -> ./backups/{timestamp}.db			
+-----+ JSON Response (HTTP 201 Created) -> Render			

6. Detailed In-Depth Pipeline Mechanics

6.1. Temporal Video Keyframe Extraction & Quota Compliance

Modern Multimodal LLM endpoints (e.g. Hugging Face Router for `Qwen2.5-VL-72B-Instruct`) enforce strict limits of **at most 5 images per request prompt**. Videos recorded in browsers or uploaded from phones often contain hundreds of frames at 30–60 FPS.

The backend implements adaptive temporal sampling in `backend/internal/ai/providers.go`:

```
// Extract 4 representative temporal keyframes across the video using ffmpeg
// Enforces max 5 image quota constraint on the multimodal inference router

cmd := exec.CommandContext(ctx, "ffmpeg", "-y", "-i", cleanPath,
    "-vf", "fps=1/2,scale=512:-1", "-vframes", "4", framePattern)

if err := cmd.Run(); err == nil {

    files, _ := filepath.Glob(filepath.Join(tmpDir, "frame*.jpg"))

    sort.Strings(files)

    for _, f := range files {

        data, _ := os.ReadFile(f)

        frames = append(frames, "data:image/jpeg;base64," +
base64.StdEncoding.EncodeToString(data))

        if len(frames) ≥ 4 { break }

    }

}
```

6.2. Unbiased Dual-Stage Clinical AI Reasoning

To ensure the AI evaluates cases on a true medical spectrum without artificial disease hallucinations or forced diagnoses:

- **Stage 1 (Vision):** Strictly extracts physical abnormalities (e.g. `"lateral recumbency, bandaged right front paw, fluid/hematochezia stain near perineum, no voluntary movement"`). No disease names are injected into the prompt.
- **Stage 2 (Veterinary Reasoning):** Evaluates evidence across infectious (viral, bacterial, parasitic), mechanical (foreign bodies, obstructions), toxicological, inflammatory, and metabolic categories using `viggovet/viggoVet-Reasoning-20B`.

💡 Unbiased Diagnostic Guidelines (System Prompt)

"1. UNBIASED EVALUATION: Formulate differentials based purely on patient species, reported complaints, and visual findings.
2. COMPREHENSIVE SPECTRUM: Systematically consider all relevant etiologies across infectious (viral, bacterial, fungal, parasitic), mechanical/surgical (obstructions, trauma), toxicological, inflammatory, and metabolic disorders.
3. EVIDENCE-BASED RANKING: Provide 4-5 differential diagnoses ranked by true clinical likelihood with pathophysiological reasoning.
4. DIAGNOSTIC WORKUP & TRIAGE: Recommend specific confirmatory lab tests, imaging modalities, and immediate emergency management."

6.3. Concurrency Resilience & Lock-Free SQLite WAL Setup

To allow concurrent read operations, write operations, and background snapshot auto-backups without `database is locked` exceptions, the SQLite connections are configured with explicit busy timeouts and single-writer pooling:

```
// Connection string with busy timeout and WAL journal mode
db, err := sql.Open("sqlite3", path + "?_foreign_keys=on&_journal_mode=WAL&_busy_timeout=10000")
db.SetMaxOpenConns(1)
db.Exec(`PRAGMA foreign_keys = ON; PRAGMA journal_mode = WAL; PRAGMA busy_timeout = 10000;`)
```

7. Database Schemas & Migration Architecture

The platform manages two segregated SQLite databases with automated versioned migrations applied

on every server startup:

Database	Table Name	Columns & Structural Schema
user.db Authentication & Security	users	id (PK), email (UNIQUE), password_hash (Bcrypt), full_name, role, created_at, updated_at
	schema_migrations	version (PK), name, applied_at
	animals	id (PK), user_id (INDEX), name, species, breed, age, sex, photo_url, notes, weight, created_at
	media	id (PK), url, path, type ('image' 'video'), mime_type, size, created_at
animal_health.db Clinical Application Data	health_screenings	id (PK), animal_id (INDEX), media_url, media_type, symptoms_json, visual_analysis_json, assessment_json, urgency (INDEX), created_at (INDEX)
	reminders	id (PK), user_id (INDEX), animal_id, type, title, description, due_at, completed, created_at
	schema_migrations	version (PK), name, applied_at

8. RESTful API Endpoint Specifications

Method & Path	Headers & Parameters	Description & Payload Structure
POST /api/auth/register	Content-Type: application/json	Registers a new user account with Bcrypt password hashing (Cost 10). Payload: { "email": "...", "password": "...", "full_name": "..." }
POST /api/auth/login	Content-Type: application/json	Authenticates credentials against user.db and issues session token. Response: { "token": "...", "user": { "id": 1, "email": "..." } }
POST /api/auth/logout	X-Auth-Token: <token>	Invalidates active user session token.

Method & Path	Headers & Parameters	Description & Payload Structure
POST /api/health-check/upload	Content-Type: multipart/form-data X-Auth-Token, X-User-ID	Uploads image or video file. Validates MIME type, stores file in ./uploads/, and registers record in media table. Response: { "id": 11, "url": "/uploads/...", "type": "video", "size": 1938155 }
POST /api/health-check/analyze	Content-Type: application/json X-Auth-Token, X-User-ID	Executes full multimodal visual inspection and veterinary differential assessment. Payload: { "media_id": 11, "symptoms": { "symptoms": [...], "other": "..." } } Response: Full structured JSON with visual analysis, assessment, differential rankings, and urgency.
GET /api/health-screenings/:id	X-Auth-Token	Retrieves complete persisted diagnostic report by ID.
GET /api/animals	X-Auth-Token	Lists all registered patients and profiles for the authenticated user.
POST /api/animals	Content-Type: application/json X-Auth-Token	Creates a new patient animal profile.
GET /api/reminders	X-Auth-Token	Lists active vaccination, medication, and clinical follow-up reminders.

9. Security Architecture & Reliability Engineering

Security & Zero-Leakage Architecture

- **Header Isolation:** User session tokens are strictly passed over X-Auth-Token and X-User-ID headers. The server's AI client injects the upstream Hugging Face API key independently via Authorization: Bearer <HUGGINGFACE_API_KEY>, preventing session token leaks and 401 collisions.
- **No Public Admin Endpoints:** All database inspection, health checks (PRAGMA integrity_check), and automated backup triggers are strictly internal to the server process and are never exposed to the public Internet.
- **Bcrypt Password Security:** User passwords in user.db are salted and hashed using adaptive Bcrypt (cost factor 10). Plaintext passwords are never stored in memory or on disk.

- **Atomic Point-in-Time Backups:** The backup engine uses native transactional `VACUUM INTO` commands, producing zero-downtime, fully verified, uncorrupted backup snapshots in `./backups/` upon every database mutation.

10. Verified Clinical Case Studies & Test Benchmarks

Test Input & Scenario	AI Vision Detection	AI Differential Diagnoses	Triage Urgency
Puppy Video WebM video clip Playful barking at kitchen bin	Alert, playful posture, voluntary movements, no skin/mucosal lesions, no respiratory distress.	1. [HIGH] Normal Puppy Play Behavior	LOW
		2. [POSSIBLE] Behavioral Response to Reflection	
		3. [LOW] Attention-seeking behavior	
Canine Neurological Motion Video Involuntary twitching, hyperkeratosis. crusting, hard pads	/ Rhythmic jaw/limb myoclonus, ocular-nasal discharge, paw pad	1. [HIGH] Canine Distemper Virus (CDV)	HIGH
		2. [MODERATE] Immune-Mediated Polyneuropathy	
		3. [POSSIBLE] Zinc-Responsive Dermatitis	
Adult Dog with Trauma dog.jpg (No symptoms supplied)	Lateral recumbency, bandaged right front paw, perineal fluid stain on floor.	1. [HIGH] Traumatic Paw Injury / Post-Op	/ MODERATE
		2. [MODERATE] Urinary Tract Infection	
		3. [POSSIBLE] Perineal Hernia / Drainage	
Critical Hemorrhagic Presentation dog.jpg + 24h bloody diarrhea vomiting	Lateral recumbency, severe prostration, continuous hematochezia fluid on floor, & bandaged leg.	1. [HIGH] Canine Parvovirus (CPV-2)	EMERGENCY
		2. [MODERATE] Acute Hemorrhagic Diarrhea Syndrome (AHDS)	
		3. [POSSIBLE] Intestinal Foreign Body Obstruction	

WAL & Multimodal AI